

Kwee–van Woerden Timing Tool

The **Kwee–van Woerden** plug-in estimates the time of an eclipse or transit minimum (or a maximum) and a timing uncertainty from a single, trimmed light-curve segment. It appears under VStar's **Tools** menu as **Kwee-van Woerden**.

Method

The time of minimum T_0 is obtained with the classic Kwee–van Woerden (1956) folding procedure: the light curve is reflected about trial mid-times, the sum of squared differences $S(T)$ is formed at an odd number of folds, and a parabola fitted to $S(T)$ yields T_0 .

The **timing error** follows Deeg (2020): after anchoring the fitted $S(T_0)$ to the photometric noise μ ,

$$\sigma^2(T_0) = 2 \mu^2 / a$$

where a is the quadratic coefficient of the $S(T)$ fit. The original 1956 error formula is also computed when it is numerically defined; on modern low-noise data (e.g. TESS) it often fails or underestimates σ . When the classic σ is undefined, it is **omitted** from the result dialog (only the Deeg σ is shown). When it is defined—as in the incomplete-eclipse example below—it appears as an extra diagnostic line.

Default folds: **5** (3 = classic KvW; 7 also available).

Workflow

1. Load photometry and switch to the **Raw Data** (JD) view.
2. Isolate **one eclipse** (filter or exclude off-eclipse points). The algorithm assumes a near-equidistant, eclipse-only segment.
3. Choose **Tools** → **Kwee-van Woerden**.
4. Select the series that contains the eclipse.
5. Set parameters (folds, initial T_1 estimate, event type, optional μ , resample).
6. Click **OK**.
7. Read $T_0 \pm \sigma$ (Deeg) from the result dialog. A compact CSV line is also copied to the clipboard when available.

For visual QC, the AAVSO “tracing paper” check is useful: reflect the light curve about T_0 and overplot the forward and reversed segments.

Parameters

Parameter	Meaning
Number of folds	3, 5 (default), or 7 reflections about the initial mid-time
Initial time estimate	Extreme magnitude (recommended) or midpoint of the segment — see below
Event type	Minimum (eclipse/transit) or maximum
Photometric noise μ	See Photometric noise below
Resample if not equidistant	Linear interpolation onto median Δt when spacing varies

Photometric noise

Deeg’s timing error needs a photometric noise μ in the **same units** as the analysed values (magnitudes or flux).

Source	When
Preferred	Off-eclipse rms measured outside the eclipse
Dialog pre-fill	If observations carry magnitude uncertainties, the field is initialised to their mean uncertainty . That is <i>not</i> the Deeg fold-based estimate—it is only a convenient starting value from the data.
Empty field	Clear the field to estimate μ from the best fold: $\mu = \sqrt{(\min(S) / (2(Z-1)))}$, assuming min S is dominated by measurement noise (Deeg). The value actually used is reported as “ μ used” in the result dialog.

You may also type any positive μ explicitly (e.g. **0.00138** for the CM Dra flux-domain demos).

Initial time estimate (T1)

The folds are built around a provisional mid-time T_1 . If T_1 is far from the true mid-eclipse, $S(T)$ may not show a usable parabolic minimum and the run can fail with errors such as:

- *Not enough $S(T)$ folds remain after symmetrisation.*
- *Parabola fit to $S(T)$ did not yield a usable minimum ($a \leq 0$).*

Choice	When to use
Extreme magnitude (default)	Usual choice for VStar: incomplete eclipses, off-centre trims, or a clear single extreme. Matches the worked examples. Prefer this when Midpoint fails with the errors above.
Midpoint of light curve	Better when the segment is well centred on the eclipse (similar ingress and egress coverage) but the extreme point is noisy or poorly defined (Deeg's preference for noisy, shape-asymmetric eclipses).

Also check the data trim: KvW expects an **eclipse-only** segment. Extra out-of-eclipse points or a window that mostly covers only ingress or only egress push Midpoint away from mid-eclipse and make failures more likely. Re-filter so the eclipse is roughly centred, or switch to Extreme magnitude.

If Extreme magnitude still fails, try fewer folds (3), ensure equidistant sampling (or leave resample on), and confirm Event type matches the data (Minimum for normal magnitudes; Maximum when the magnitude column is really normalised flux that dips at mid-eclipse, as in the appendices).

Worked example 1: complete eclipse (CM Dra epoch 7024)

This example uses the complete primary eclipse of CM Dra at epoch 7024 from [hdeeg/KvW example_data](#) (`CMDra7024.lc`). The GitHub `.lc` file is whitespace-separated and is **not** readable as-is by VStar's Download or Simple observation sources. Use the converted CSV in [Appendix A](#) instead.

Steps in VStar

1. Choose **File** → **New Star from File...** → **Download or Simple** → **Request Observation Text**.
2. Copy the CSV block from [Appendix A](#) and paste it into the observation text box, then load the star.
3. Ensure you are in the **Raw Data** view. The series is already eclipse-only.
4. **Tools** → **Kwee-van Woerden**, select the loaded series.
5. Parameters:
 - Folds: **5 (recommended)**
 - Initial time estimate: **Extreme magnitude (recommended)**
 - Event type: **Maximum** — Appendix A places **normalised flux** in the magnitude column (lower at mid-eclipse). Choosing Maximum makes the algorithm treat that dip correctly when the column is read as a VStar magnitude field. For ordinary AAVSO magnitudes (larger = fainter at mid-eclipse), use **Minimum** instead.
 - Photometric noise μ : enter **0.00138** to match the Deeg/README flux-domain demo, or leave empty to estimate.
 - Resample: can be left on; this fixture is already equidistant.
6. Click **OK**.

7. Confirm T_0 is near **58739.9291169** (BJD–2400000, as in the pasted times) and σ (Deeg) $\approx$ **0.0000125** d ($\approx$ 1.1 s). Classic KvW σ is omitted here (undefined for this light curve).

Worked example 2: incomplete eclipse (CM Dra epoch 7023)

This uses the incomplete primary at epoch 7023 (`CMDra7023.1c`), which Deeg uses to illustrate unbalanced coverage and S(T) branch cropping. Paste the CSV from [Appendix B](#).

Steps in VStar

1. Load Appendix B the same way as Appendix A (**File** → **New Star from File...** → **Download or Simple** → **Request Observation Text**).
2. **Tools** → **Kwee-van Woerden** with the same parameter choices as example 1 (5 folds, Extreme magnitude (recommended), **Maximum**, $\mu = 0.00138$). Midpoint of light curve often fails on this incomplete segment—the geometric mid-time of the data is not near mid-eclipse.
3. Click **OK**.
4. Confirm $T_0 \approx 58738.6607358$, σ (Deeg) ≈ 0.0000191 d. For this light curve the classic KvW σ is defined and should appear as approximately **0.0000662** d.

Worked example 3: AAVSO AID photometry (MR Aps)

This walkthrough uses real AAVSO International Database (AID) observations of the southern eclipsing binary **MR Aps** (VSX type EB, period $\approx$ 0.528 d).

Load the data

1. Choose **File** → **New Star from AAVSO Database...** (or the equivalent AID menu item in your VStar build).
2. Star name: **MR Aps**
3. Julian Date range: **2460796** to **2460798**
4. Load the observations (leave only Johnson V selected).

This JD window contains a **single eclipse**.

Prepare the series

1. Stay in the **Raw Data** (JD) view.
2. Select the Johnson V series.

Run Kwee–van Woerden

1. **Tools** → **Kwee-van Woerden**, select the series.
2. Suggested parameters for AID magnitudes:
 - Folds: **5 (recommended)**
 - Initial time estimate: **Extreme magnitude (recommended)**
 - Event type: **Minimum (eclipse / transit)** — AID magnitudes are fainter (larger) at mid-eclipse
 - Photometric noise μ : leave empty to estimate from min S, keep the mean magnitude-uncertainty pre-fill if present, or enter a typical uncertainty for the series if known
 - Resample: on if the cadence is uneven

Kwee-van Woerden parameters

Number of folds
5 (recommended)

Initial time estimate
Extreme magnitude

Event type
Minimum (eclipse / transit)

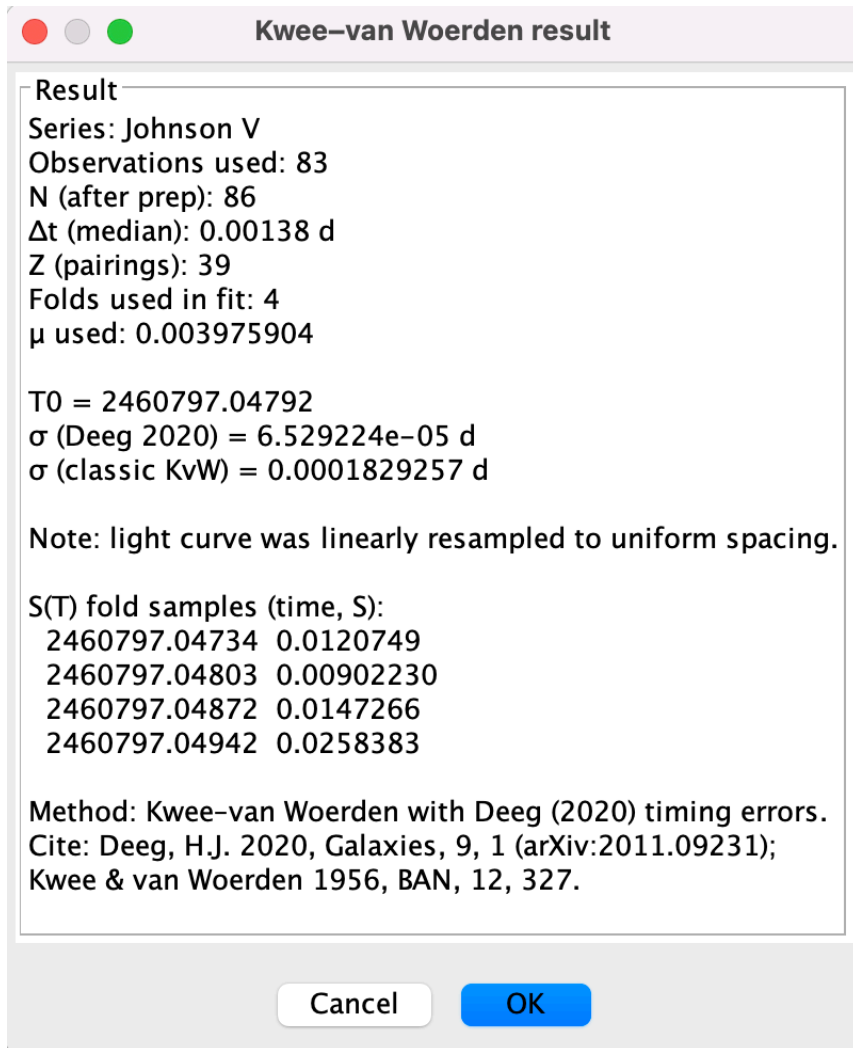
Photometric noise μ (empty = estimate)
0.003975903614

☒ Resample if not equidistant

Help Cancel OK

1. Click **OK**.
2. Confirm results near:

Quantity	Value
T_0	2460797.04792
σ (Deeg 2020)	6.529224×10^{-5} d
σ (classic KvW)	0.0001829257 d



Limitations

- Input should be **one eclipse**, not a long multi-cycle light curve.
- Strongly **asymmetric** eclipses violate the symmetry assumption.
- Large gaps or highly uneven sampling need resampling or pre-interpolation.
- μ must be in the **same units** as the values analysed.
- A poorly centred or one-sided segment combined with **Midpoint** T_1 often produces S(T) fit failures; see [Initial time estimate \(T1\)](#).

Attribution

This plug-in is a **Java implementation** based on the Deeg (2020/2021) paper, not a transpile of the [hdeeg/KvW](#) Python/IDL code. That repository was used for golden-value verification and to clarify procedural details not fully specified in the paper (in particular half-integer fold pairing and S(T) branch cropping for incomplete eclipses).

References

- Kwee, K.K. & van Woerden, H. 1956, Bull. Astron. Inst. Netherlands, 12, 327.
- Deeg, H.J. 2020, “A Modified Kwee–Van Woerden Method for Eclipse Minimum Timing with Reliable Error Estimates”, *Galaxies*, 9, 1. [arXiv:2011.09231](https://arxiv.org/abs/2011.09231) · [ADS](https://ui.adsabs.org/2020Galax...9...1D)
- Reference code: <https://github.com/hdeeg/KvW>
- AAVSO guidance on times of minimum: [Determining the Time of Minimum](https://www.aavso.org/faq/faq-10-determining-the-time-of-minimum)

Appendix A: CM Dra epoch 7024 converted to CSV

Source: https://github.com/hdeeg/KvW/blob/main/example_data/CMDra7024.lc

Copy everything below (including the comment lines) and paste it into the text box opened by **File** → **New Star from File...** → **Download or Simple** → **Request Observation Text**.

```
#lightcurve of primary eclipse of CM Dra at epoch 7024
# BJD-TDB,      norm. flux
58739.90842904,0.9472268
58739.90981790,0.9282367
58739.91120677,0.9027383
58739.91259563,0.8765095
58739.91398450,0.8457783
58739.91537336,0.8197890
58739.91676222,0.7847447
58739.91815109,0.7523733
58739.91953995,0.7204994
58739.92092881,0.6859992
58739.92231768,0.6516519
58739.92370654,0.6194330
58739.92509541,0.5881648
58739.92648427,0.5583567
58739.92787313,0.5329423
58739.92926200,0.5253384
58739.93065086,0.5385398
58739.93203973,0.5633745
58739.93342859,0.5947506
58739.93481745,0.6270559
58739.93620632,0.6588940
58739.93759518,0.6948199
58739.93898405,0.7258901
58739.94037291,0.7601001
58739.94176177,0.7934963
58739.94315064,0.8235733
58739.94453950,0.8524282
58739.94592836,0.8814062
58739.94731723,0.9084927
58739.94870609,0.9325403
```

Appendix B: CM Dra epoch 7023 converted to CSV

Source: https://github.com/hdeeg/KvW/blob/main/example_data/CMDra7023.lc

Incomplete primary eclipse (ingress-heavy). Paste as for Appendix A.

```
#lightcurve of primary eclipse of CM Dra at epoch 7023
# BJD-TDB,      norm. flux
58738.65150805,0.7110910
58738.65289691,0.6779842
58738.65428578,0.6422326
58738.65567464,0.6104053
58738.65706350,0.5786916
58738.65845236,0.5521469
58738.65984122,0.5289302
58738.66123009,0.5262330
58738.66261895,0.5436970
58738.66400781,0.5722702
58738.66539667,0.6013605
58738.66678553,0.6352366
58738.66817440,0.6680585
58738.66956326,0.7037288
58738.67095212,0.7349917
58738.67234098,0.7687948
58738.67372985,0.8026004
58738.67511871,0.8316808
58738.67650757,0.8600320
58738.67789643,0.8872918
58738.67928529,0.9154191
58738.68067416,0.9374043
```